A Dual-Tank Model for Real-Time Tracking of Phosphocreatine and Glycolytic Energy Reserves in Cycling

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Abstract

Endurance-cycling analytics have converged on the Critical Power (CP) / W′ framework, in which all work performed above a sustainable power asymptote is drawn from a single finite reserve, W′. This lumps two physiologically distinct anaerobic systems — the fast phosphocreatine (PCr / alactic) system and the slower glycolytic (lactic) system — into one tank with a single recovery time constant. That simplification is convenient but discards the information a rider most wants during hard, variable-intensity efforts: *which* reserve is being spent, and *how fast* each will come back. This paper proposes a **dual-tank model** that splits W′ into a fast PCr reserve and a slow glycolytic reserve, drives both from the instantaneous power trace, and applies system-specific depletion and restoration laws drawn from the bioenergetic and 31P-MRS literature. The model is deliberately reduced to be computable at 1 Hz on a wrist- or bar-mounted device, and we specify a Garmin Connect IQ implementation that displays live reserve, consumption, and depletion for both systems. We close with a validation strategy and an explicit statement of the model's assumptions and limits.

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1. Introduction

A power meter reports mechanical output, not metabolic cost. The bridge between the two is a model of the body's energy-supply systems. The dominant bridge in cycling — the CP/W′ model — treats supra-CP work as depletion of a single reservoir and sub-CP work as its exponential refill (Skiba et al. 2012). It is power-native, validated, and computationally trivial, which is why it underlies W′bal displays in GoldenCheetah, intervals.icu, and several head units.

Its central simplification is also its central weakness for real-time decision-making: **W′ is one tank.** Physiologically, anaerobic capacity is at least two tanks with very different dynamics (Margaria; di Prampero & Ferretti 1999):

- The **phosphocreatine (PCr / alactic)** system — high power, low capacity, **fast** oxidative recovery (seconds to a minute), but with recovery that is pH-sensitive and slows across repeated bouts.

- The **glycolytic (lactic)** system — larger capacity, lower peak power, and **slow** recovery (minutes) that effectively only proceeds at low intensity.

A rider deciding whether to cover the next surge cares which tank is empty. If the PCr tank is full but glycolytic is depleted, a 5-second jump is fine but a 40-second dig is not — a distinction the single-tank W′bal cannot express. The goal of this paper is a model that keeps the power-native, on-device virtues of W′bal while restoring the two-system resolution.

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2. Physiological background

Phosphocreatine (alactic). ATP is buffered by the creatine-kinase reaction (PCr + ADP → ATP + Cr). Intramuscular PCr (~20–25 mmol·kg^{-1} wet muscle) is the highest-power energy source and is drawn on instantly at any power transient. Its resynthesis is oxidative and fast: classic alactic O_2-debt half-time ≈ 30 s (Margaria); 31P-MRS gives recovery time constants τ_PCr ≈ 20–40 s, strongly **pH-dependent** (acidosis slows it) and **biphasic** under high glycolytic load (Harris et al. 1976; Yoshida et al. 2013, PMID 23662804). Recovery also **slows progressively across repeated hard bouts.**

Glycolytic (lactic). Anaerobic glycolysis converts glycogen to lactate, supplying large ATP flux for ~10 s–2 min. Its "fuel level" is best tracked as accumulated muscle/blood lactate — a proxy for H^+ and metabolite accumulation. Contribution decays across repeated sprints (≈40% of ATP in the first of 10×6 s efforts, <10% by the last; Sci Rep 2024, DOI 10.1038/s41598-024-78916-z). Restoration is slow (lactic O_2-debt half-time ≈ 15 min) and, mechanistically, proceeds fastest at low intensity, effectively ceasing above the first lactate threshold (LT1).

The onset that matters. Both systems contribute from the start of a hard effort, but not equally at first: PCr is drawn on instantly (it is the highest-power, immediate buffer), while glycolytic flux takes several seconds to ramp up as glycogen phosphorylase is activated (Parolin et al. 1999) — so the *early* demand is PCr-weighted, shifting toward a shared draw as glycolysis engages. This graded onset — and the very different refill rates — is what the dual-tank model must reproduce (it does so with a parallel, PCr-weighted draw and a glycolytic activation ramp; §4.2), not a strict "PCr first, then glycolytic" hand-off.

What the two "tanks" represent — and what they don't. The "tank" is a deliberate simplification for intuition and on-device display, not a claim that either system is a literal reservoir that drains. The two sides are not even the same *kind* of quantity:

- The **PCr tank** does correspond to a real, depletable substrate store — intramuscular phosphocreatine — so "PCr empty" is close to literal.
- The **glycolytic "tank"** is better understood as **tolerance to accumulating fatigue-related metabolites** than as a fuel gauge. On the ~1–4 min timescale it represents, muscle glycogen is nowhere near exhausted; carbohydrate depletion is a separate, hours-long limiter this model does not address. What forces power down at the limit of a hard effort is the buildup of inorganic phosphate

(Pi), H^+ (acidosis), ADP, and extracellular K^+ toward the limit of tolerance — consistent with exhaustion at CP/W′ coinciding with a reproducible low-PCr / high-metabolite state. (Lactate itself is largely a fuel and a marker, not the cause of force loss.)

This is the physiological reason the two recovery laws differ: PCr resynthesizes quickly, whereas the glycolytic side recovers slowly and only at low intensity because it reflects *clearance and buffering* of accumulated byproducts, which need aerobic metabolism and time — exactly why the model tracks the glycolytic state as an accumulation variable (a muscle-lactate proxy; §4.1) and gates its recovery below LT1. Muscle fatigue is multifactorial and still debated; W′ deliberately lumps it into one number for usability. In short: **the PCr tank is a fuel that depletes; the glycolytic "tank" is a byproduct bucket that fills.**

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3. Prior models (and why a reduced dual-tank sits between them)

Three model families exist (full survey in `literature-review-anaerobic-models.md`):

- **Family A — CP / W′-balance** (Skiba 2012/2015; Bartram 2018; Froncioni–Clarke). Power-native and on-device-friendly, but **one lumped tank, single recovery τ.** Recovery `τ_W′ = 546·e^(−0.01·D_CP) + 316 s` cannot represent fast PCr + slow glycolytic simultaneously.
- **Family B — hydraulic tank models** (Morton 1986; Weigend `three_comp_hyd` 2021). **Two explicit anaerobic vessels** (AnF = phosphagen, AnS = glycolytic) plus an aerobic vessel; out-predicts W′bal for intermittent recovery (arXiv 2108.04510). But its 8 parameters are abstract and fit per-athlete by evolutionary computation — heavy for an embedded target.
- **Family C — bioenergetic supply/demand ODEs** (EJAP 2023, PMID 37369795). Explicit alactic / lactic / aerobic metabolic-rate terms, with an efficiency-limited PCr recovery and a power-gated lactate-removal law. Mechanistically ideal but has 17 parameters and needs grey-box fitting.

The gap: Family A is implementable but under-resolved; Families B and C resolve the two systems but are too heavy to run and calibrate on a head unit. The dual-tank model below keeps the two-tank resolution of B/C with the on-device simplicity of A.

To be precise about what "lighter" means here, because it is easy to oversell: the model's core parameter count (`CP, W′, f_p, P_p_max, τ_p, τ_g, LT1, η, τ_on`, ~9, plus two optional realism terms) is essentially the **same** as the hydraulic model's eight — it is **not** a smaller model. The difference is *where the parameters come from*. Only `CP` and `W′` are fitted per athlete (from a standard CP test); the rest are **hard-coded to literature defaults** rather than fit by evolutionary computation. That buys on-device simplicity and a trivial calibration, at a real cost: the load-bearing defaults (`f_p`, `P_p_max`, `LT1`, and the recovery τ's) are exactly the quantities a CP test does **not** determine, so the model's two-system resolution rests on assumptions, not measurements. That is a legitimate identifiability-for-convenience trade — but it is a trade, and the sections below are explicit about which numbers are earned and which are assumed.

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4. The dual-tank model

4.1 State and parameters

Two energy reserves (Joules), each with a fixed capacity:

- `R_p` — **PCr reserve**, capacity `C_p`
- `R_g` — **glycolytic reserve**, capacity `C_g`
- Total anaerobic capacity `W′ = C_p + C_g`

Rider-supplied parameters (all obtainable from a maximal power–duration / CP test):

Symbol	Meaning	Typical default
`CP`	critical power (W)	from 3-/12-min CP test (≈ FTP + a few %)
`W′`	total work above CP (J)	from same test (~15–25 kJ)
`f_p`	fraction of W′ assigned to the PCr tank	**0.25** (weakly identified — see §7)
`P_p_max`	PCr peak (immediate) power above CP (W)	≈ P_{1s} − CP (best 1 s power − CP)
`g_pmax`	glycolytic peak power, as a fraction of `P_p_max`	0.5· `P_p_max` (PCr is the higher-power system)
`τ_p`	PCr recovery time constant (s)	22 (fast, pH-modulated)
`τ_g`	glycolytic recovery time constant (s)	360 (slow)
`LT1`	first lactate threshold power (W); recovery of the glycolytic tank proceeds below it	**measured** (a threshold test), *not* a fixed %CP
`η`	PCr recovery-rate efficiency (see note)	0.80

Derived: `C_p = f_p·W′`, `C_g = (1 − f_p)·W′`. Initial state `R_p = C_p`, `R_g = C_g` (full).

Three of these deserve flags up front, because the defaults are doing real work:

- **`f_p` is weakly identified, not measured.** It is *not* determined by a CP test. It is nudged by the reconstitution curve (§6.3) and by physiological alactic-fraction estimates, both of which put it **below** the 0.35 we used previously — hence the revised 0.25 default. It should be personalized.
- **`P_p_max` ≈ P_{1s} − CP** (not a 5 s figure): at the very first second glycolysis is barely active (`g ≈ 1 − e^(−1/6) ≈ 15%`), so the instantaneous ceiling is close to, but slightly above, CP + `P_p_max` — i.e. P_{1s} − CP mildly over-attributes to PCr and should be read as an upper bound.
- **`η` is a recovery-*rate* efficiency, not hysteresis.** In the recovery law below it is mathematically equivalent to scaling `τ_p` to `τ_p/η` (the asymptote is still full recovery, so there is no path dependence). It is retained only as an interpretable knob mapping to the aerobic-efficiency term of the

EJAP 2023 model, and is degenerate with `τ_p` when fitted — the two cannot be identified separately. A *true* efficiency loss (incomplete recovery under fatigue) would require an asymptote **below** `C_p`; that is a possible extension, not what `η` does here.

4.2 The 1 Hz update

Let `P` be instantaneous power (W) and `Δt = 1 s`. Define the demand relative to aerobic supply as `Δ = P − CP`.

Case 1 — depletion (`Δ > 0`, above CP). Demand `need = Δ·Δt` (J) is met by **both systems in parallel**. This matters: glycolytic ATP supply is not instantaneous. Glycogen phosphorylase (the rate-limiting glycolytic enzyme) transforms to its active form over the first several seconds of maximal effort — Parolin *et al.* (1999) measured it rising from 12% at rest to 47% by 6 s — while phosphocreatine is the immediate buffer, near-fully drawn within the first ~10 s (Bogdanis *et al.* 1996); both contribute from the onset, not in sequence (González-Alonso *et al.* 2000). We model this with a **glycolytic activation** term `g ∈ [0,1]` that ramps first-order with a time constant `τ_on ≈ 6 s` while above CP and relaxes back during recovery:

```
g ← g + (1 − g)·(1 − exp(−Δt/τ_on))      # activation ("glycolytic inertia"); decays b
g_pmax = 0.5·P_p_max                      # glycolytic peak rate < PCr peak rate

# Rate-proportional split: demand shared in proportion to each system's available rate
total_rate = P_p_max + g_pmax·g
share_p = need · P_p_max / total_rate      # at g=0 PCr covers all; at g=1 a ~2:1 PCr:
share_g = need − share_p

take_p = min(share_p, R_p, P_p_max·Δt)     # both systems are rate-capped AND capacity
take_g = min(share_g, R_g, g_pmax·Δt)
unmet  = need − take_p − take_g
# any shortfall (a tank empty or rate-capped) spills to the partner, then to deficit:
take_g += min(unmet, R_g − take_g, g_pmax·Δt − take_g);   unmet −= …
take_p += min(unmet, R_p − take_p, P_p_max·Δt − take_p);  unmet −= …
R_p −= take_p;  R_g −= take_g

if unmet > 0:  exhaustion = true          # both tanks empty/capped → rider at/over th
```

At effort onset `g ≈ 0`, so PCr supplies essentially everything (the fast transient); as `g → 1` over a few seconds the two tanks drain together — but *not* equally: because PCr is the higher-power system (`g_pmax = 0.5·P_p_max`), the steady-state split is roughly **2:1 in PCr's favour**, and the glycolytic tank is rate-limited to `g_pmax` so it cannot absorb an arbitrarily large one-second demand. Once the small PCr tank empties, glycolysis carries the rest up to its own cap.

On identifiability (and consistency with §7). The parallel draw is the physiologically faithful choice for the *live display*, and `τ_on` is a literature-set constant (Parolin 1999), not a fitted one. But we are careful not to over-claim: the depletion split is **not** identifiable from power at all — total draw above CP is all the pedals see, and any (`f_p`, `g_pmax`) that sums to the same total is observationally equivalent. In *principle* the bi-exponential **recovery** of W′ constrains the split (a fast PCr component + a slow glycolytic component); in *practice* that fit is ill-conditioned (§6.3 shows the fast component is invisible at standard reconstitution sampling), so `f_p` ends up **assumed and weakly constrained**, not recovered. This is the §7 position, and it is the one to trust — the reserves are a plausible decomposition, not measured latent state.

Case 2 — restoration (`Δ ≤ 0`, at/below CP). Recovery "headroom" is `CP − P`. Each tank refills exponentially toward full; glycolytic refill is gated by intensity.

```
# PCr: fast, oxidative (η rescales the effective rate to τ_p/η — see §4.1 note)
R_p += η · (C_p − R_p) · (1 − exp(−Δt/τ_p))

# Glycolytic: slow, only meaningfully below LT1 (a MEASURED power, not a %CP)
if P < LT1:
    gate = (LT1 − P) / LT1                              # 0 at LT1, → 1 toward zero power
    R_g += gate · (C_g − R_g) · (1 − exp(−Δt/τ_g))
```

`LT1` is the rider's **first lactate threshold in watts**, from a threshold test — *not* a fixed fraction of CP. LT1 and CP vary independently (LT1 is roughly 65–85% of CP depending on training status), and this gate decides whether the glycolytic tank refills at all during tempo/endurance riding, so getting it wrong there is consequential. Where only CP is known, `LT1 ≈ 0.80·CP` is a fallback, but it is the weakest default in the model and should be replaced by a real measurement.

Optional realism (recommended, off by default): - *pH-slowed / fatiguing PCr recovery:* scale `τ_p` upward as the glycolytic tank is emptier, e.g. `τ_p_eff = τ_p · (1 + k·(1 − R_g/C_g))`, capturing the observed slowing across repeated bouts. **-** *Aerobic ramp:* replace the hard `CP` boundary with a first-order aerobic supply `A(t)` that rises toward `min(P, CP)` with `τ_aer ≈ 25 s`, and use `need = (P − A)·Δt`. This reproduces the onset "oxygen deficit" that transiently loads the anaerobic tanks even below CP.

4.3 Reported quantities

- **PCr reserve** `= R_p / C_p` (0–100%) — how much "punch" is left.
- **Glycolytic reserve** `= R_g / C_g` (0–100%) — how much "sustained dig" is left.
- **Consumption rate** (per system, W) — `take_p/Δt`, `take_g/Δt`, the live draw on each tank.
- **Combined W′bal** `= (R_p + R_g)/W′` — backward-compatible with existing single-tank displays.
- **Depletion / exhaustion flag** — both reserves at/near zero.

4.4 Why this behaves correctly

- A short, very hard surge draws mostly from `R_p` (glycolysis has not yet ramped in), and `R_p` visibly refills within ~30–60 s of easy pedaling — matching PCr physiology.
- A sustained supra-CP effort empties `R_p` quickly, then leans on `R_g`, which only comes back over minutes and only when the rider drops below LT1 — matching glycolytic physiology.
- **On "generalizing W′bal" — a careful claim.** In *depletion* the model is a faithful generalization: as long as no tank is empty or rate-capped, `take_p + take_g = need`, so the summed draw above CP matches standard W′bal exactly. In *recovery* it is deliberately **different**: bi-exponential, with an LT1 gate and the `η` / `τ_p` rate — it does **not** reduce to Skiba's single `τ_W′ = 546·e^(−0.01·D_CP) + 316`. Nor does `f_p → 0` recover standard W′bal: it collapses to a *single glycolytic tank* with `τ_g` and an LT1 gate, which is still not the incumbent's recovery law. So this is a generalization in **structure and depletion behaviour**, not a strict superset that contains W′bal as one parameter setting. The divergence is intentional — the whole point is that single-tank recovery is the thing we are trying to improve on.

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5. Real-time on-device implementation (Garmin Connect IQ)

The model's per-second cost is a handful of multiplies and one `exp()` per tank — trivial for a head unit. A companion document, `connectiq-app-spec-and-prompt.md`, gives the full technical specification and a ready-to-use build prompt. Summary of the target design:

- **App type:** a custom full-screen **Data Field** (`Toybox.WatchUi.DataField`), which the head unit calls once per second — a natural fit for the `Δt = 1 s` update.
- **Input:** `Activity.Info.currentPower` inside `compute(info)`.
- **Configuration:** `CP`, `W′`, `f_p`, and the recovery constants exposed via Connect IQ app settings (`Application.Properties`), so a rider enters them from Garmin Connect.
- **Display:** two vertical bar gauges (PCr and glycolytic reserve) with numeric % and color bands (green → amber → red), plus optional live consumption in W.
- **Recording:** write both reserves and consumption to the FIT file via `FitContributor` fields so they sync to Garmin Connect / intervals.icu / Strava for post-ride analysis.
- **Footprint:** two `Float` state variables, no arrays, well within Connect IQ memory budgets.

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6. Validation strategy

The model is a hypothesis; it must be checked against data before its numbers are trusted. The honest difficulty is that the model's *headline feature* — a separate PCr reserve tracked in real cycling — is the hardest thing to validate, so we separate tests that only check the plumbing from the one test that would actually earn the second tank.

1. **Face validity / unit tests** — synthetic power traces (single sprint, repeated sprints, supra-CP hold, sub-CP recovery) should produce the qualitative behaviours in §4.4.
2. **Backward compatibility — *depletion only*.** Summing the tanks reproduces a reference W′bal implementation (Froncioni–Clarke) on the same trace **in depletion** (`take_p + take_g = need`). It will **not** match in recovery, by construction (§4.4), so this test must be scoped to above-CP segments — a test that expected agreement everywhere is one the model is designed to fail.
3. **Reconstitution targets — with two caveats.** After a full W′ depletion, combined recovery can be compared to the published curve (≈37% / 65% / 86% at 2 / 6 / 15 min; Chorley & Lamb 2020). But: (a) **this curve is blind to the fast tank.** A `τ_p ≈ 22 s` process is `1 − e^(−120/22) ≈ 99.6%` complete by the first (2 min) sample, so passing this test validates the *glycolytic* recovery and the *size* of the PCr offset — **not** the PCr recovery kinetics, i.e. not the feature that distinguishes this model from W′bal. To constrain `τ_p` you need recovery samples at ~10/20/30/45/60 s (test 5). (b) **our defaults do not hit it, and that is informative.** At the old `f_p = 0.35` the model recovers ~53% by 2 min against the 37% target — a 16-point overshoot — because putting 35% of W′ in a tank that is ~fully back by 2 min forces combined 2-min recovery above 35%. Matching the curve pulls `f_p` **down** (~0.13 at passive rest, ~0.25 at a realistic soft-pedal recovery power); this is the main reason the default was revised to 0.25. Note also that a single slow exponential cannot fit all three points exactly (the empirical curve is itself multi-exponential), so this is a *soft constraint on* `f_p`, not a pass/fail on the decomposition.
4. **Fast-recovery kinetics (the missing piece).** The load-bearing test for `τ_p` is a repeated-bout protocol with **early** recovery sampling — e.g. all-out efforts separated by 10/20/30/45/60 s — and the between-bout power recovery correlated with modelled PCr recovery (as Bogdanis 1996 did for PCr). Standard reconstitution data cannot supply this; it must be measured deliberately.
5. **System-specific truth (lab) — and its hard limit.** Blood-lactate kinetics can corroborate the glycolytic tank, but the PCr tank's gold standard, **31P-MRS, is essentially unavailable in real cycling**: the measurement requires the muscle to be inside a magnet, so it exists only for small-muscle knee-extension or immediate post-exercise, not whole-body cycling. The model's headline novelty therefore **cannot be checked against the gold standard in the modality it targets** — a limitation to state plainly, not paper over.
6. **The test that earns the second tank — out-of-sample intermittent tolerance.** Because the model is a near-superset of single-tank in depletion, it can *always* fit at least as well, so "it fits better" is guaranteed and is **not** evidence the compartments are real. The decisive test is a **head-to-head against single-tank W′bal on an out-of-sample, intermittent-effort tolerance prediction** — the same protocol class on which the hydraulic model beat W′bal (arXiv 2108.04510): predict time-to-exhaustion / whether a prescribed interval set is completable, where the dual-tank and single-tank models make *different* predictions, and show the dual-tank wins. Until that test is passed, the second tank is a plausible, physiologically-motivated decomposition — not a validated one.
7. **Field calibration** — fit `f_p`, `P_p_max`, and the `τ`'s per athlete to maximal-effort tests (a sprint-then-hold or repeated-bout protocol) rather than trusting defaults.

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7. Assumptions and limitations

- **The PCr/glycolytic split `f_p` is assumed and weakly constrained, not measured.** Power alone cannot identify the depletion split (§4.2); recovery can in principle, but the fast component is invisible at standard sampling (§6.3), so `f_p` is effectively an assumption. Independent lines (reconstitution offset ~0.13–0.25; physiological alactic-fraction estimates ~0.20–0.30; some multi-ride calibrations higher) put it in a broad ~0.15–0.35 band. We default to 0.25 and treat it as a per-athlete calibration target — the model's outputs are sensitive to it.
- **The reserves are a decomposition, not latent state.** The two bars are a physiologically-motivated split of one measured quantity (W′bal), not two independently measured reserves. When the PCr tank is full — which, given `τ_p ≈ 22 s`, is most of the time except the ~30–60 s after a hard effort — the glycolytic bar is an affine transform of the existing single W′bal (`W′bal = f_p + (1−f_p)·R_g/C_g`), so it carries no new information there. The PCr bar adds decision-relevant content **only** in those post-surge transients. Those transients *are* where race decisions concentrate (can I cover this attack right after the last one?), which is the case for the display — but the honest framing is a heuristic decomposition whose split is assumed, not "the two numbers a rider races on" as if both were measured.
- **`η` is degenerate with `τ_p`** (§4.1): as written it rescales the recovery rate, not a true efficiency, and the two cannot be fitted separately. Reported `η` / `τ_p` pairs are non-unique.
- **`LT1` should be measured, not derived from CP.** The `0.80·CP` fallback will be wrong for many riders (LT1 ranges ~65–85% of CP), and it gates whether the glycolytic tank recovers during tempo, so a bad value materially changes recovery estimates.
- **The headline feature is hard to validate in-modality.** 31P-MRS cannot be run during real cycling (§6.5), and depletion "fit-better" is guaranteed by construction, so the second tank is only earned by an out-of-sample intermittent-tolerance head-to-head vs single-tank (§6.6) — not yet performed here.
- **Fixed time constants** ignore the documented pH-dependence and bout-to-bout slowing of PCr recovery unless the optional fatigue term is enabled.
- **Hard CP boundary** omits the aerobic slow component and onset kinetics unless the optional aerobic-ramp term is enabled; expect over-attribution to anaerobic tanks in long low-intensity recovery (the same failure mode reported for the EJAP 2023 model).
- **Power is whole-body and mechanical**, not muscle-specific; the model cannot see local fatigue, cadence, or fiber-type effects.
- **Not medical or diagnostic.** This is a training/pacing aid; tank levels are estimates, not measurements of muscle chemistry.
- **Parameters drift** with fitness, fatigue, heat, and altitude; periodic re-testing is required.

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8. Conclusion

The single-tank W′ model earned its place by being power-native and light enough to run anywhere, but it answers "how much anaerobic work is left?" without answering "in which system?" The dual-tank model proposed here offers a physiologically-motivated split at negligible computational cost: divide W′ into a fast, rate-limited, fast-recovering PCr tank and a slow, large, slowly-recovering glycolytic tank; drain them **in parallel** from the power trace (PCr-weighted, with glycolysis ramping in); and refill them with system-specific, intensity-gated laws. It generalizes W′bal in depletion and diverges from it — deliberately — in recovery, it runs at 1 Hz on a Garmin head unit, and it surfaces a two-system decomposition — *punch left* and *dig left* — that is most informative in the transients where pacing decisions concentrate.

Two honest caveats set the agenda. The split `f_p` is **assumed, not measured**, and the display's extra resolution over single-tank W′bal is real but concentrated in post-surge windows. And the decomposition is not yet **validated**: the tests that only check the plumbing (backward compatibility, reconstitution offset) it can pass, but the one that would earn the second tank — an out-of-sample intermittent-tolerance win over single-tank W′bal — remains to be run, and the PCr gold standard (31P-MRS) is unavailable in real cycling. The accompanying Connect IQ specification turns the model into a live data field; the open questions are calibration (`f_p`, `LT1`, per-athlete τ) and that decisive head-to-head field validation.

References

See `docs/literature-review-anaerobic-models.md` for the full annotated bibliography with DOIs and PMIDs. Primary sources for this paper:

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Sourcing note: equations for the incumbent models were cross-checked against PubMed Central full text (PMC7552657, PMC10638188); typeset equations there were image-embedded, so published closed forms were reconstructed from the surviving prose and standard literature. Verify exact coefficients against publisher PDFs before hard-coding.